02561 – COMPUTER GRAPHICS

Project Report

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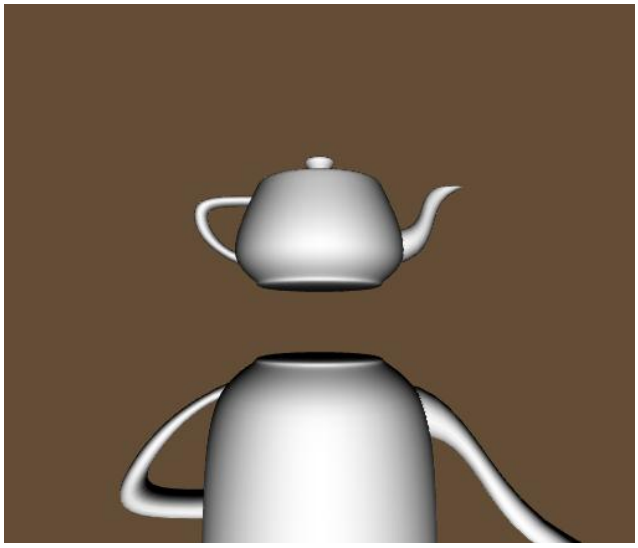
Project website link + Worksheet & Exercises: https://jonatanrasmussen.github.io/



IIII Introduction

This project is inspired by *"Project Initiator 1: Planar Reflector"*

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The ability to depict reflections is important for creating immersive and visually impressive scenes. This project, undertaken as part of the class 02561 Computer Graphics at DTU Compute, will explore the implementation of planar reflectors using the WebGL framework.

Planar reflectors are surfaces that can be mathematically defined by a single plane. This simple defintion makes rendering a somewhat simple process, as the "reflection" is simulated by mirroring the vertices of an object across the defined plane. To demonstrate this concept, the project utilizes a teapot, a standard model in computer graphics, placed in front of a reflective planar surface.

The purpose of the project is to understand and create a reflection transformation matrix that facilitates the rendering of the teapot's mirror image, paired with the appropriate reflection of the lighting to maintain visual consistency. Together, this gives the "illusion" of a reflection, even though it is simply two individually rendered teapots. To make the project more nuanced, the teapot will not remain static; instead, it will hover slowly up and down over the plane, and the mirrored object will obviously have to follow this movement, but in inverse relative to the reflective plane.

IIII Methodology

The methodology for this project is straightforward, as it is leaning heavily on the technical aspects outlined in the core literature of computer graphics that has been covered throughout the course. The project will be carried out using the WebGL framework to render the teapot and its planar

reflection. The approach was divided into two phases; rendering the reflected object and achieving a degree of realism through blending and clipping techniques.

The setup involves creating a WebGL scene with the stencil buffer enabled and setting a camera with a 65-degree field of view positioned at coordinates (0, 0, 1), with the look-at point set to (0, 0, −3). This provided a foundational perspective from which the reflection phenomena could be observed and assessed. The specific values were copied from the teacher's Project Initiator document.

The rendering of the reflective teapot model utilized a reflection transformation matrix R. The matrix was derived mathematically to facilitate the mirroring of the teapot's vertices across the planar reflective surface — in this case, a ground quad. The matrix R must be constructed to specifically invert the teapot across the reflective plane, similar to a mirror, or an ice lake.

IIII Implementation

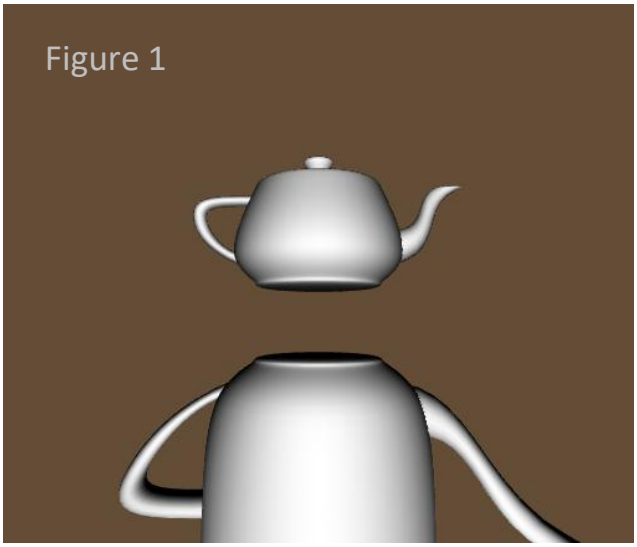
As mentioned in the methodology section, the matrix R must be constructed to reflect and translate the teapot across the reflective plane so that it appears under the ground plane. The following image documents and explains its implementation:

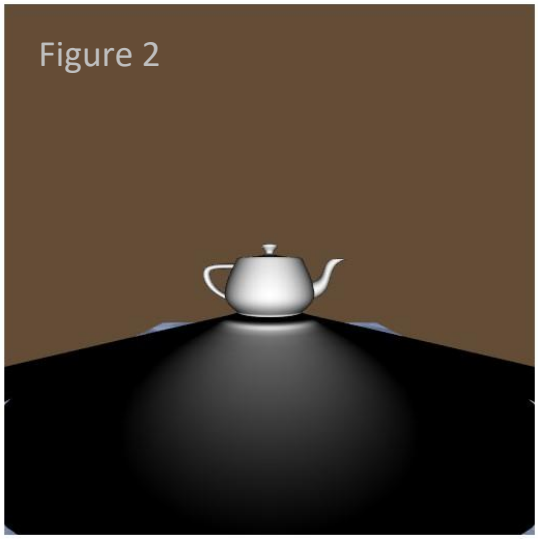
```
// The reflection matrix should not only reflect the teapot but also translate it
// so that it appears under the ground plane
let reflectedModelView = mult(teapotModelView, mat4(
    1, 0, 0, 0,
    0, -1, 0, 0, // Flips the teapot over the ground plane
    0, 0, 1, 0,
    0, distanceAboveGroundPlaneY, 0, 1 // Translates teapot to appear under ground plane
));
```

The scene was a continuation of the work done in week 9's worksheet. The camera was set to a 65 degrees field of view and placed in (0, 0, 1) looking at (0, 0, −3). The rendering of the ground plane was temporarily disabled (commented out, as stated in the project initiator), and the teapots' / ground planes' y-coordinate was adjusted to match. The next section will cover the results.

IIII Results and Discussion

The reflection was not implemented correctly and this negatively impacted the results. The end result was a mirrored and translated teapot rather than a convincing reflection, as the ground plane has been commented out. Re-inserting the ground plane would cover the “reflection”, as it did not properly project onto the plane. The implementation of the reflection transformation matrix R was only partially successful – it should be noted however that mirroring across the plane was achieved. Figure 1 display the “mirrored” teapot, without a visible ground plane. Figure 2 displays the scene with the ground plane present, but the teapot is distorted in a way that does not accurately projects onto it. Further and better results were not pursued.





The project's aspiration to render a semi-transparent ground plane that blended the reflection encountered technical obstacles that could not be overcome. Despite this, the journey of rendering planar reflections using WebGL was explored and documented. The results were far from desirable, but nevertheless presented and conveyed the fundamental principles of mirroring objects across a plane and dynamically adjusting the reflection based on object movement. The current scene is a stepping stone to more complex reflection rendering in future work, which can hopefully achieve the full and convincing planar reflection described in the project description.